

Experiment 9

Problem Statement

You are required to design and execute an experiment to compare infrastructure isolation and automation mechanisms using **container-based environments** and **virtual machines**. Use **Docker** and **VirtualBox** to perform the tasks and analyze their differences.

Task 1: Set Up Containerized Environment

1. Create a project directory.
 2. Develop a simple web application (index.html) displaying a message indicating it is running inside a container.
 3. Create a Dockerfile to:
 - Use a web server base image
 - Copy the application file into the container
 - Expose the required port
 4. Build a Docker image and run a container from it.
 5. Access the application through a web browser and verify its execution.
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Task 2: Set Up Virtualized Environment

1. Create a virtual machine using **VirtualBox**.
2. Install a Linux operating system (e.g., Ubuntu) in the VM.
3. Inside the VM:

- Install a web server
 - Deploy a webpage displaying a message indicating it is running inside a virtual machine
4. Access the application using the VM's IP address and verify execution.
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Task 3: Analyze Infrastructure Isolation

1. Observe and compare:
 - Process-level isolation in containers
 - OS-level isolation in virtual machines
 2. Record differences in:
 - Resource usage
 - Startup time
 - Environment separation
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Task 4: Automate Container Deployment

1. Remove the running container.
 2. Re-deploy the application using a single command.
 3. Observe and document how automation simplifies deployment.
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Task 5: Automate Virtual Machine Environment

1. Use snapshot or cloning features in **VirtualBox**.

2. Restore or replicate the configured VM environment.
 3. Compare the automation effort and time with container deployment.
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Task 6: Comparative Evaluation

1. Create a comparison table between container and VM environments based on:
 - Isolation level
 - Resource utilization
 - Startup time
 - Deployment speed
 - Portability
 2. Analyze which approach is more suitable for modern DevOps practices.
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Task 7: Analytical Questions

1. Explain how **Docker** provides infrastructure isolation.
2. Analyze how isolation in **VirtualBox** differs from containers.
3. Evaluate the advantages and limitations of containerization over virtualization.
4. How does automation differ between containers and virtual machines?
5. Recommend an approach for large-scale deployment and justify your answer.

Expected Outcome

- The application should run successfully in both container and VM environments.
- Clear differences between container-based and VM-based isolation should be observed.
- Automation in container environments should be faster and more efficient compared to virtual machines.

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain answers/snapshots of outputs corresponding to each task.
3. The PDF name should be in the form "StudentName_SAPID".